LAB 4 – SQL FUNCTIONS

Conversion Functions

```
CREATE TABLE employees (
employee_id INT PRIMARY KEY,
first_name VARCHAR(50),
last_name VARCHAR(50),     email
VARCHAR(100) UNIQUE,
hire_date DATE,
department_id INT );

INSERT INTO employees VALUES
    (201, 'Likith', 'Kumar', 'likith.k@company.com', '2020-02-14', 10),
    (202, 'Maanas', 'J', 'maanas.j@company.com', '2019-06-22', 20),
    (203, 'Hemanth', 'Gowda', 'hemanth.g@company.com', '2021-09-08', 30),
    (204, 'Madhan', 'Kumar', 'madhan.k@company.com', '2018-12-17', 40),
    (205, 'Aditya', 'C', 'aditya.c@company.com', '2022-03-25', 20),
    (206, 'Dhavan', 'Ganesh', 'dhavan.g@company.com', '2020-10-11', 30);

SELECT * FROM employees;
```

employee_id	first_name	last_name	email	hire_date	department_id
201	Likith	Kumar	likith.k@company.com	2020-02-14	10
202	Maanas	J	maanas.j@company.com	2019-06-22	20
203	Hemanth	Gowda	hemanth.g@company.com	2021-09-08	30
204	Madhan	Kumar	madhan.k@company.com	2018-12-17	40
205	Aditya	C	aditya.c@company.com	2022-03-25	20
206	Dhavan	Ganesh	dhavan.g@company.com	2020-10-11	30

Implicit Data Type Conversion

```
SELECT employee_id, first_name, department_id
FROM employees
WHERE department_id > 20;
```

employee_id	first_name	department_id
203	Hemanth	30
204	Madhan	40
206	Dhavan	30

Explicit Data Type Conversion

```
SELECT employee_id, first_name,
       CAST(hire_date AS CHAR) AS hire_date_text
FROM employees;
```

employee_id	first_name	hire_date_text
201	Likith	2020-02-14
202	Maanas	2019-06-22
203	Hemanth	2021-09-08
204	Madhan	2018-12-17
205	Aditya	2022-03-25
206	Dhavan	2020-10-11

Date Functions

```
SELECT     first_name,
last_name,
hire_date,
CURDATE() AS today,
    YEAR(hire_date) AS year,
    MONTH(hire_date) AS month,
    DAY(hire_date) AS day,
    YEAR(CURDATE()) - YEAR(hire_date) AS approximate_years
FROM employees;
```

first_name	last_name	hire_date	today	year	month	day	approx_years
Likith	Kumar	2020-02-14	2026-11-01	2020	2	14	6
Maanas	J	2019-06-22	2026-11-01	2019	6	22	7
Hemanth	Gowda	2021-09-08	2026-11-01	2021	9	8	5
Madhan	Kumar	2018-12-17	2026-11-01	2018	12	17	8
Aditya	C	2022-03-25	2026-11-01	2022	3	25	4
Dhavan	Ganesh	2020-10-11	2026-11-01	2020	10	11	6

LAB 5 – SQL OPERATORS

```
SELECT     employee_id,     first_name,     department_id,
employee_id + 10 AS id_plus_10,     employee_id > 203 AS
id_greater_than_203,     department_id = 20 AND employee_id > 203 AS
dept20_and_id_above_203,     department_id = 20 OR department_id = 30 AS
dept20_or_dept30,     NOT department_id = 20 AS not_dept20
FROM employees;
```

employee_id	first_name	department_id	id_plus_10	id_gt_203	dept20 AND id>203	dept20 OR 30	NOT dept20
201	Likith	10	211	0	0	0	1
202	Maanas	20	212	0	0	1	0
203	Hemanth	30	213	0	0	1	1
204	Madhan	40	214	1	0	0	1
205	Aditya	20	215	1	1	1	0
206	Dhavan	30	216	1	0	1	1

LAB 6 – SPECIAL OPERATORS & SET OPERATIONS

Special Operators

```
SELECT employee_id, first_name, department_id, email
FROM employees
WHERE employee_id BETWEEN 202 AND 205
   OR department_id IN (10, 30)
   OR first_name LIKE 'M%'
   OR department_id IS NULL;
```

employee_id	first_name	department_id	email
201	Likith	10	likith.k@company.com
202	Maanas	20	maanas.j@company.com
203	Hemanth	30	hemanth.g@company.com
204	Madhan	40	madhan.k@company.com
205	Aditya	20	aditya.c@company.com
206	Dhavan	30	dhavan.g@company.com

UNION

```
SELECT first_name
FROM employees
WHERE department_id = 10

UNION

SELECT first_name
FROM employees
WHERE department_id = 30;
```

first_name
Dhavan
Hemanth
Likith

INTERSECT

```
SELECT department_id
FROM employees
WHERE employee_id <= 203

INTERSECT

SELECT department_id
FROM employees
WHERE employee_id >= 203;
```

department_id
20
30